Repeated Games: A Folk Theorem

For a stage game $G = (N, \{A_i\}_{i \in N}, \{\pi_i\}_{i \in N})$ and $i \in N$, define $V_i : A_{-i} \to \mathbb{R} \cup \{\infty\}$ by

$$V_i(a_{-i}) = \sup_{a_i \in A_i} \pi_i(a_i, a_{-i}).$$

Let $\pi = (\pi_1, ..., \pi_n)$.

Theorem (*Folk Theorem I: SPNE and "Nash reversion"*)**.** *Let $\tilde{a} \in NE(G)$, and let $a^* \in A$ satisfy*

- $\pi(a^*) \gg \pi(\tilde{a})$*; and*
- *for each $i \in N$, $V_i(a^*_{-i}) < \infty$.*

Then there is a $\delta^* \in (0,1)$ such that, if $\delta \in (\delta^*, 1)$, then $h^*_\infty = h_0 a^* \cdots a^* \cdots$ is a SPNE ***outcome*** *of $G^\infty(\delta)$.*

This is the simplest of the Folk Theorems and it barely scratches the surface, even for complete information games of almost-perfect information. (*Almost-perfect* in the sense that each stage game is a simultaneous-move game, and so the game itself is not a perfect-information game; but otherwise each player knows the history.) An important extension are games of *imperfect monitoring*, meaning that one or more players simply observe a noisy statistic of past actions. For more on repeated games, I recommend Mailath and Samuelson's (2006) *Repeated Games and Reputations*.